

1. Administrative & Study Identification

Posting ID: 72273127606077281 **Study Title:** Efficacy and Safety of Remibrutinib After Switching From Ocrelizumab in Participants Living With Relapsing Multiple Sclerosis. **Full Official Study Title:** A Randomized, Open-label, Parallel-group, Non-inferiority Study Comparing Efficacy, Safety, and Tolerability of Remibrutinib After Switching From Ocrelizumab in Participants Living With Relapsing Multiple Sclerosis, Followed by Open-label Treatment With Remibrutinib **Short Title/Acronym:** RESHAPE Study **Protocol Number:** CLOU064C12306 **NCT Number:** NCT06846281 **Trial Status:** Recruiting **Sponsor Name / Company:** Novartis **Investigational Product:** Remibrutinib **Phase of Development:** Phase 3 (Phase 3b) **Phase Criteria Type:** PHASE_REQUIREMENT **Phase Criteria Text:** Trial must be in PHASE3 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy, safety, and tolerability of remibrutinib after switching from ocrelizumab and compared to continuous ocrelizumab treatment, in patients living with relapsing multiple sclerosis (plwRMS). **Secondary Objectives:** To evaluate the proportion of participants with no evidence of disease activity-3 (NEDA-3) and to monitor the number of adverse events (AEs) and serious adverse events (SAEs).

2.2 Background & Rationale To address the real-world clinical question of whether patients doing well on anti-CD20 therapy (ocrelizumab) might benefit from switching to a Bruton's tyrosine kinase (BTK) inhibitor (remibrutinib). This is particularly relevant for aging populations living with MS who have accumulating concerns about prolonged B-cell depletion, cumulative immunosuppression, and associated infection risks.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Ocrelizumab) **Blinding:** Open-label **Randomization:** Randomized (1:1), Parallel-group, Non-inferiority

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 360 participants **Number of Sites:** Global, across roughly 20 countries (approx. 94 locations including the USA) **Estimated Study Start:** 23-Jul-2025 **Estimated Primary Completion:** 20-Jul-2029

4. Subject Selection (Eligibility Criteria)

4.1 Key Inclusion Criteria 1. Male or female aged 40 to 70 years (inclusive). 2. Diagnosis of Relapsing Multiple Sclerosis (RMS) according to the 2017 McDonald diagnostic criteria. 3. Treated with ocrelizumab according to routine clinical practice and at standard dose. 4. Neurologically stable within 30 days prior to randomization. 5. Suitable to be switched to remibrutinib based on physician judgement or patient preference.

4.2 Key Exclusion Criteria 1. Diagnosis of primary progressive multiple sclerosis (PPMS) according to the revised 2017 McDonald criteria. 2. History of clinically significant Central Nervous System disease or neurological disorders. 3. History of confirmed Progressive Multifocal Leukoencephalopathy (PML) or neurological symptoms consistent with PML. 4. Active clinically significant systemic bacterial, viral, parasitic, or fungal infections. 5. Active, chronic disease of the immune system other than MS. 6. Severe cardiac disease or significant findings on the ECG. 7. Participant who is unable to undergo MRI scans. 8. History of life-threatening infusion or injection reaction related to ocrelizumab.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - *Experimental Arm:* Remibrutinib tablet taken orally. - *Active Comparator Arm:* Ocrelizumab at standard dose. - **Drug Name:** ocrelizumab - **Trade Names:** Ocrevus - **ATC Code:** L04AA36 - **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/ocrevus-epar-product-information_en.pdf **Route of Administration:** Oral (remibrutinib) / Intravenous or Subcutaneous (ocrelizumab). **Duration of Treatment:** Core Part (CP) maximum duration of up to 24 months, followed by an optional Extension Part (EP) of up to 24 months of open-label, single-arm, fixed-dose remibrutinib for eligible participants.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for symptomatic management of MS and related co-morbidities. **Prohibited:** Concurrent immunosuppressive therapies or experimental disease-modifying therapies (DMTs) outside of the protocol-assigned arms.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -30 to 0 | Informed Consent, Medical History, Neurological Exam, baseline MRI | | **Baseline** | Day 0 | Randomization (1:1), First Dose (Oral Remibrutinib or IV/SC Ocrelizumab) | | **Interim (Core)** | Months 1 to 24 | Safety Labs,

Efficacy Assessment (NEDA-3 tracking), MRI Scans, AE Monitoring |
| **End of Study** | Month 48 | Completion of 24-month Extension
Part, Final MRI, Exit Interview, IP Accountability |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized rate of new or enlarging T2 lesions_Core Part: Number of new/enlarging T2 lesions per year on MRI at month 24 (relative to baseline).

Measurement Method: Central Magnetic Resonance Imaging (MRI) assessment from Baseline up to Month 24.

7.2 Secondary & Exploratory Endpoints 1. Proportion of participants with no evidence of disease activity-3 (NEDA-3)_Core Part. 2. Number of Adverse Events (AEs) and Serious Adverse Events (SAEs)_Core Part. 3. Annualized rate of new or enlarging T2 lesions_Extension Part.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient reporting, clinical visits, and laboratory/MRI monitoring assessing for infection risks and disease stability. **Serious**

Adverse Events (SAEs): Standard protocol; reported within typically 24 hours of site awareness. **Data Monitoring Committee (DMC):** No DMC is instituted for this study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (non-inferiority) outcomes; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Target $N = 360$ to adequately power the trial to demonstrate non-inferiority of remibrutinib against ocrelizumab. **Handling of Missing Data:** Standard protocol statistical methodologies, utilizing Multiple Imputation (MI) and model-based estimators.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring of trial sites to ensure protocol adherence. **Audit Trail:** Detailed tracking via eCRF locking and centralized blinded readings of MRI endpoints.

11. Study Identifiers & Governance

Primary ID: CLOU064C12306 **Registry Number:** NCT06846281 **Sponsor/Lead Organization:** Novartis **Official Regulatory Title:** A Randomized, Open-label, Parallel-group, Non-inferiority Study Comparing Efficacy, Safety, and Tolerability of Remibrutinib After Switching From Ocrelizumab in Participants Living With Relapsing Multiple Sclerosis, Followed by Open-label Treatment With Remibrutinib.

12. Clinical Framework (Multiple Sclerosis Specific)

Disease Area: Relapsing Multiple Sclerosis **Preferred UMLS Name:** Multiple Sclerosis, Relapsing-Remitting **Disease Preferred Name:** Multiple sclerosis relapse **Disease Definition:** Relapsing Multiple Sclerosis is an autoimmune disease of the central nervous system characterized by unpredictable attacks of new or worsening neurological symptoms, known as relapses or exacerbations, which are followed by periods of partial or complete recovery called remissions. During remission, symptoms may improve or stabilize, but some residual disability can accumulate over time. It is the most common form of multiple sclerosis. **MeSH Code:** C08.381.560.675 **SNOMED ID:** 247000007 **SNOMED Hierarchy:** 247000007|Multiple sclerosis|Multiple sclerosis (disorder) **Diagnosis Threshold:** 2017 McDonald diagnostic criteria **Medical Methodology:** Transition from anti-CD20 therapy to Bruton's Tyrosine Kinase (BTK) inhibitor **Optimization Target:** Sustained disease control (measured via T2 lesions/NEDA-3) with potentially improved long-term safety profile regarding infection risk

13. Study Procedures & Methodology

Management Pathway: Step-down/transition protocol from complete B-cell depletion (ocrelizumab) to selective BTK inhibition (remibrutinib). **Education Protocol (HCP):** Evaluation of patient suitability for switching, monitoring for PML and systemic infections. **Education Protocol (Patient):** Counseling on treatment switch and adherence to daily oral tablet (experimental arm). **Quality Audit Mechanism:** Centralized evaluation of MRI scans and neurological stability.

14. Observation & Follow-Up Schedule

Total Duration: Up to 48 months (24 months Core Part + 24 months Extension Part) **Onsite Visit Cadence:** Pre-defined milestones (e.g., Months 6, 12, 18, 24) aligned with clinical and MRI assessments. **Remote Monitoring Frequency:** Routine telemedicine or remote symptom tracking check-ins based on local site protocol.

Checklist Review Interval: Regular reviews of AE profiles, lab results, and MS symptom exacerbations at scheduled checkpoints.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				